FBS Embryology — NBME Study Guide

Gametogenesis · Menstrual cycle · Fertilization to implantation · Weeks 2–3 · Neurulation & body cavities · Placenta · Prenatal testing & twins (includes Anki-deck embryology)

Red = extra high yield (classic NBME) Black = core content *Italic blue = mnemonic*

1. Meiosis & chromosome counting

Cell	Ploidy (n)	DNA (c)
Spermatogonium / oogonium (before S)	2n	2c
Primary spermatocyte / oocyte (after S)	**2n (46)**	**4c**
Secondary spermatocyte / oocyte (after meiosis I)	**1n (23)**	**2c**
Spermatid / ovum (after meiosis II)	1n	1c

- **Meiosis I = reductional (homologs separate; crossing over in prophase I). Meiosis II = equational (sister chromatids separate).**
- **Nondisjunction (mostly maternal meiosis I) ↑ with maternal age → trisomy 21, 18, 13; Klinefelter (47,XXY), Turner (45,X).**
- Primordial germ cells arise in the **yolk sac wall (epiblast origin)** and migrate to the gonadal ridge by week 5–6; lost PGCs → germ cell tumors (sacrococcygeal teratoma).

2. Spermatogenesis & the male tract

- Begins at puberty, in seminiferous tubules; ~64–74 days. Spermatogonia (mitosis) → primary spermatocyte → secondary spermatocyte → spermatids → **spermiogenesis** (shape change only, no division).
- **Spermiogenesis: acrosome from Golgi (hyaluronidase, acrosin), flagellum from centriole, mitochondria form midpiece sheath, cytoplasm shed.**
- **Sertoli cells (FSH)**: blood–testis barrier (tight junctions), androgen-binding protein, **inhibin B** (→ ↓ FSH), **Müllerian-inhibiting factor/AMH** (Müllerian ducts regress), support spermatogenesis. **Leydig cells (LH)**: interstitial, make **testosterone** *(L for LH, L for Leydig)*.
- Sperm path: *SEVEN UP* — Seminiferous tubules → (rete testis, efferent ductules) → Epididymis (maturation, motility, storage) → Vas deferens → Ejaculatory duct → (Nothing) → Urethra → Penis.
- **Capacitation** (in female tract; removal of cholesterol/glycoproteins) → **acrosome reaction** on binding ZP3 of the zona pellucida.

3. Oogenesis & the ovary

- **Oogonia multiply only before birth (months 3–7)**; primary oocytes **arrest in prophase I (diplotene) until puberty**; most undergo **atresia** (~7 million fetal → 1–2 million at birth → ~400,000 at puberty → ~400 ovulated).
- Each cycle: LH surge → meiosis I completes → secondary oocyte + 1st polar body → **secondary oocyte arrested in metaphase II until fertilization** *("Ovulated, waiting to Meet a sperm")*.
- Follicle stages: primordial (flat cells) → **primary** (single cuboidal layer; zona pellucida forms) → secondary (2+ layers, antrum begins) → tertiary/**Graafian** (large antrum, cumulus oophorus).
- **Two-cell model: theca cells (LH) make androgens → diffuse to granulosa cells (FSH) → aromatase → estrogen.** *Your deck:* theca interna = vascular, hormone-secreting layer (estrogen); theca externa = fibrous.
- Granulosa = female counterpart of Sertoli; theca = counterpart of Leydig.
- **Corpus luteum** (from ruptured follicle) → **progesterone** (+ estrogen). Degenerates after ~14 days → corpus albicans, unless **hCG** from the syncytiotrophoblast rescues it (until placenta takes over ~weeks 8–10).

4. Menstrual cycle & hormones

Phase	Days	Ovary	Endometrium	Hormones
Menstrual	1–5	Early follicle recruitment	Functional layer shed	↓ estrogen & progesterone

Follicular / proliferative	~6–14	Dominant follicle grows	Proliferates (estrogen)	FSH ↑ early, estrogen rises
Ovulation	**~14**	**Graafian follicle ruptures 24–36 h after LH surge**	**—**	**Estrogen peak → positive feedback → LH surge**
Luteal / secretory	**15–28 (fixed 14 days)**	**Corpus luteum**	**Glands secrete glycogen; receptive**	**Progesterone ↑ (basal body temp ↑ ~0.5 °F)**

- **Variation in cycle length comes from the follicular phase; luteal phase is fixed ~14 days.**
- **Menses occurs because the corpus luteum degenerates → progesterone falls → spiral arteries constrict → ischemia → shedding.**
- Estrogen: endometrial proliferation, thin stretchy cervical mucus (spinnbarkeit). Progesterone: secretory endometrium, thick mucus, maintains pregnancy, ↑ temperature.
- **Combined OCPs**: negative feedback ↓ FSH/LH → no follicle maturation, no LH surge (no ovulation); thick cervical mucus; thin endometrium.
- Endometrium layers: functional (shed) vs basal (regenerates). **Decidua** = pregnancy endometrium: basalis (maternal placenta), capsularis, parietalis.

5. Fertilization, cleavage & implantation (week 1)

- **Fertilization in the ampulla of the uterine tube, within ~24 h of ovulation.**
- Sperm penetrate corona radiata → **zona pellucida** (ZP3 binding, acrosome reaction) → fusion. **Cortical reaction / zona reaction blocks polyspermy** (Ca^{2+} wave; ZP changes). Polyspermy → triploidy (partial mole 69,XXY).
- **Results of fertilization: meiosis II completed, diploidy restored, sex determined (X or Y sperm), cleavage initiated.**
- Cleavage: mitosis without growth → blastomeres → **morula (16 cells, ~day 3)** → enters uterus → **blastocyst (~day 4–5)**: outer cells pump Na^+ in, water follows → blastocyst cavity. **Trophoblast** (→ placenta) + **inner cell mass/embryoblast** (→ embryo).
- Hatches from zona → **implantation begins ~day 6** in the posterior-superior uterine wall (endometrium in secretory/luteal phase). hCG detectable in blood ~day 8, urine ~day 10–14.
- **Ectopic pregnancy**: most often **ampulla**; risk ↑ with PID/salpingitis, prior ectopic, surgery, IUD, Kartagener. Abdominal pain + vaginal bleeding + ↑ hCG, empty uterus on US. Placenta previa = implantation near/over the internal os (painless bleeding).

6. Week 2: “week of twos”

- **2 layers of trophoblast: cytotrophoblast (inner, mitotic) + syncytiotrophoblast (outer, invasive, makes hCG).**
- **2 layers of embryoblast (bilaminar disc): epiblast (→ amniotic cavity; later gives all 3 germ layers) + hypoblast (→ primary yolk sac).** *Your deck* pairs epiblast ≈ ectoderm, hypoblast ≈ endoderm.
- 2 cavities: amniotic cavity & yolk sac (+ chorionic cavity from splitting of extraembryonic mesoderm). Connecting stalk → umbilical cord.
- **Villi**: primary (cytotrophoblast core + syncytio) → secondary (+ extraembryonic mesoderm core) → **tertiary (+ fetal blood vessels, end of week 3)**.
- **hCG** (α subunit like LH/FSH/TSH) maintains the corpus luteum; very high in moles/choriocarcinoma (can cause hyperthyroidism, hyperemesis).
- **Hydatidiform mole**: complete = 46,XX (empty egg + paternal duplication), no fetus, “snowstorm”/grape-like, very ↑ hCG, ↑ choriocarcinoma risk. Partial = 69,XXY (2 sperm), fetal parts present, lower risk.

7. Yolk sac, primitive gut & umbilical cord

- Yolk sac: early blood formation (blood islands, week 3), primordial germ cells, contributes to primitive gut (folding incorporates it).
- **Vitelline duct normally obliterates; persistence → Meckel diverticulum** (rule of 2s: 2% of people, 2 feet from ileocecal valve, 2 inches long, 2 types of ectopic tissue — gastric/pancreatic, presents by age 2; painless bleeding) or vitelline fistula (meconium from umbilicus).

- **Umbilical cord: 2 arteries (deoxygenated blood to placenta) + 1 vein (oxygenated to fetus), Wharton jelly.** Single umbilical artery → associated with renal and cardiac anomalies.
- **Allantois** → urachus (median umbilical ligament); patent urachus → urine from umbilicus.

8. Week 3: gastrulation & germ layers

- **Gastrulation = formation of 3 germ layers from epiblast via the primitive streak (begins with primitive node/streak at the caudal end).** Epiblast cells invaginate: first replace hypoblast → **endoderm**; then between → **mesoderm**; remaining epiblast → **ectoderm**.
- **Notochord** (from primitive node) → induces neural plate; adult remnant = **nucleus pulposus**.
- **Sacrococcygeal teratoma** = remnant of primitive streak (most common tumor in newborns). **Caudal regression syndrome** (sirenomelia) — maternal diabetes.
- Oropharyngeal (buccopharyngeal) and cloacal membranes: ectoderm + endoderm, no mesoderm.
- **Week 3–8 = embryonic period: most susceptible to teratogens (organogenesis).** Weeks 1–2: "all-or-none".

Germ layer	Gives rise to
Surface ectoderm	Epidermis, hair, nails, lens, anterior pituitary (Rathke pouch), enamel, mammary/sweat glands, inner ear
Neuroectoderm	**Brain, spinal cord, retina, posterior pituitary, pineal, oligodendrocytes, astrocytes**
Neural crest	**PNS ganglia, Schwann cells, melanocytes, adrenal medulla (chromaffin), odontoblasts, parafollicular C cells, pia/arachnoid, aorticopulmonary septum, craniofacial bones *(MOTEL PASS)***
Mesoderm	**Muscle, bone, connective tissue, heart, blood/lymph vessels, kidneys, gonads, dermis, spleen, adrenal cortex, serous linings**
Endoderm	**Gut epithelium, liver, pancreas, lungs, thyroid follicular cells, thymus, parathyroids, bladder/urethra epithelium**

9. Mesoderm, body cavities & neurulation

- Mesoderm divisions: **paraxial** → somites (sclerotome = vertebrae/ribs; myotome = skeletal muscle; dermatome = dermis); **intermediate** → urogenital system (kidneys, gonads); **lateral plate** → somatic (parietal) layer (body wall, limb bones/CT, parietal serosa) & splanchnic (visceral) layer (gut wall muscle, heart, visceral serosa).
- **Intraembryonic coelom** (space within lateral plate) → pericardial, pleural and peritoneal cavities; septum transversum → central tendon of diaphragm, part of liver.
- **Neurulation (weeks 3–4)**: notochord induces neural plate → neural folds → neural tube; **anterior neuropore closes ~day 25, posterior ~day 27–28**.
- **Neural tube defects**: anencephaly (anterior neuropore; polyhydramnios — can't swallow), spina bifida occulta (no herniation, tuft of hair/dimple), meningocele, myelomeningocele (posterior neuropore). **↑ AFP in maternal serum & amniotic fluid, ↑ AChE in amniotic fluid**. Prevent: **folic acid**. Risk: valproate, carbamazepine, maternal diabetes, folate deficiency.

10. Placenta: structure, variants & pathology

- Fetal part = **chorion frondosum** (villous chorion); maternal part = **decidua basalis**. Exchange across the **placental membrane** (syncytiotrophoblast, cytotrophoblast — thins later — villous connective tissue, fetal endothelium). Maternal blood fills intervillous spaces from spiral arteries.
- **Syncytiotrophoblast makes hCG, hPL (human placental lactogen → insulin resistance), estrogen (estriol), progesterone.**
- Crosses: O_2, CO_2, glucose, **IgG (only immunoglobulin)**, lipid-soluble drugs, alcohol, many viruses (TORCH), anti-D antibodies (Rh disease). Does not cross: IgM, heparin, insulin (largely).
- **Placenta accreta / increta / percreta**: defective decidua basalis → villi attach to / invade / penetrate myometrium; risk prior C-section; postpartum hemorrhage.
- **Placental abruption**: premature separation → **painful** bleeding (cocaine, smoking, HTN, trauma); **Placenta previa** → **painless** bleeding.

- **Preeclampsia**: abnormal spiral artery remodeling → HTN + proteinuria after 20 weeks; eclampsia = + seizures (Mg sulfate); HELLP.
- Variants: succenturiate (accessory) lobe, velamentous cord insertion (vasa previa risk), battledore (marginal), circumvallate.

11. Prenatal testing, amniotic fluid & twins

- **Amniotic fluid**: mostly fetal urine after ~week 16; removed by fetal swallowing. Normal ~1 L at term.
- **Polyhydramnios (>2 L)**: can't swallow — esophageal/duodenal atresia, anencephaly; maternal diabetes, twins. **Oligohydramnios (<0.5 L)**: can't make/excrete urine — bilateral renal agenesis, posterior urethral valves, placental insufficiency, ROM → **Potter sequence** (pulmonary hypoplasia, flat face, limb deformities, clubfoot).
- **Screening**: quad screen (~15–20 weeks): AFP, hCG, estriol, inhibin A. **Down syndrome: ↓ AFP, ↑ hCG, ↓ estriol, ↑ inhibin A**. Trisomy 18: all ↓ (inhibin normal). NTD/abdominal wall defect/twins: ↑ AFP. First-trimester: ↑ nuchal translucency, ↓ PAPP-A, ↑ β-hCG (Down). Cell-free fetal DNA from ~10 weeks.
- **Diagnostic**: **chorionic villus sampling (~10–13 weeks)**; **amniocentesis (~15–20 weeks)**; percutaneous umbilical blood sampling (cordocentesis). Lecithin/sphingomyelin ratio >2 = fetal lung maturity.
- **Twins**: dizygotic (fraternal, 2/3; two eggs) → always dichorionic diamniotic. **Monozygotic (identical) — timing of split decides membranes**:

Split (days)	Stage	Result
0–4	Morula (2-cell to morula)	Dichorionic, diamniotic (2 placentas)
4–8	**Blastocyst (inner cell mass splits)**	**Monochorionic, diamniotic (most common MZ) — twin-twin transfusion risk**
8–12	Bilaminar disc	Monochorionic, monoamniotic (cord entanglement)
>13	**Embryonic disc**	**Conjoined twins**

12. Rapid-fire NBME pearls

- **Primary oocyte: prophase I arrest until puberty. Secondary oocyte: metaphase II arrest until fertilization.**
- **LH → Leydig (testosterone); FSH → Sertoli (ABP, inhibin, AMH). Theca androgens → granulosa estrogen.**
- **Fertilization in ampulla; implantation day 6; ectopic usually ampulla.**
- **Week 2 = 2s (cyto/syncytio; epi/hypoblast; amnion/yolk sac). Week 3 = 3 germ layers (gastrulation), notochord → nucleus pulposus.**
- **NTD: ↑ AFP + ↑ AChE; prevent with folate. Down: ↓ AFP, ↑ hCG, ↓ estriol, ↑ inhibin.**
- **Complete mole 46,XX paternal, no fetus; partial 69,XXY with fetal parts.**
- **Monochorionic diamniotic = split days 4–8 (most common MZ twins).**
- **Oligohydramnios → Potter sequence (renal agenesis); polyhydramnios → GI atresia/anencephaly.**